Flávio Silva

Junior Software Engineer · Python · Full-stack · Data & LLMs

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SUMMARY

Master's student in Informatics Engineering at the University of Minho, finishing a dissertation on using large language models to generate data quality checks. Builds and ships software end to end, from C and Python to TypeScript front ends, with automated tests, CI/CD, Docker and Kubernetes. Looking for a junior software engineering role or a professional internship in the Braga area.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL, Haskell
Frameworks: FastAPI, Node.js (Express), React, React Native, Vue.js, Next.js
Data: PostgreSQL, SQLite, ClickHouse
DevOps & tools: Docker, Kubernetes, Helm, GitHub Actions (CI/CD), Linux, Git, pytest
AI: LLM APIs (Gemini, Claude), local models with Ollama

EDUCATION

MSc in Informatics Engineering, University of Minho, Braga · 2024 – Nov 2026
BSc in Informatics Engineering, University of Minho, Braga · 2020 – 2024

PROJECTS

MSc Dissertation: LLM framework for data quality · 2025 – 2026 · *Python, SQL, ClickHouse, LLMs*

- Framework, developed with Jscrambler, in which a large language model proposes data quality tests from each table's context (schema, descriptions, transformation SQL and lineage).
- Every proposal is checked by deterministic validators against the schema and the real data before it becomes SQL; a human expert makes the final decision.
- Evaluated on a real ClickHouse pipeline against the data engineers' own tests: asking the model what each SQL transformation guarantees raised the share of their tests found from 39% to 67%.

JobRadar: open-source job finder for Portugal · 2026 · *Python, FastAPI, SQLite, Vue 3, Android, GitHub Actions*

- Reads 30+ job boards and company careers sites (APIs, feeds and web pages), keeps the junior jobs and internships that fit the user's area and level, and scores each one against their profile.
- Optional AI (Gemini or Claude) explains each score, drafts application messages, tailors the CV and writes follow-ups.
- Ships as a Windows installer, one-line installers for macOS and Linux, and an Android app running the same Python code; 221 automated tests, and every release is installed and tested on an Android emulator and on Windows by GitHub Actions. github.com/FuzzyLaDuzzy/jobradar

F3M: medication care app · 2025 · *React Native (Expo), Node.js/Express, TypeScript, PostgreSQL (Prisma)*

- Mobile app that helps patients with mild to moderate dementia manage their medication, built for the Informatics Project course in collaboration with the company F3M.
- Built REST API endpoints and the relational database that let caregivers manage medication schedules, prescriptions and pill reminders, with an alarm and notification system. github.com/MartimRedondo/F3M_APP

PictuRAS: image editor · 2025 · *Vue.js, REST API, Docker, Kubernetes, Helm*

- Web application to edit and showcase images with filters applied in the browser: Vue.js front end, backend API and email testing with Mailhog, deployed with Docker on Kubernetes (Minikube) and packaged with Helm. github.com/JoaoCoelho2003/PictuRas

Client/Server Task System · 2023 · *C, Linux*

- Server that runs programs for many concurrent clients through pipes and forks, reports scheduled and finished tasks with their run times, and uses a custom "mysystem" built with fork and exec.

LANGUAGES

Portuguese (native) · English (fluent)